

Risk Measures under Paired-Ambiguity: A Deep Learning Reflected BSDE Framework

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Abstract

We study optimal stopping under dynamic risk measures with simultaneous ambiguity in the probability model and the discount rate. We introduce a paired ambiguity framework combining Girsanov model uncertainty with cash subadditive risk evaluation and characterize the stopping value by an upper reflected backward stochastic differential equation (BSDE). We establish structural properties of the resulting stopping operator and study quadratic drivers associated with entropic risk measures, obtaining explicit stopping rules in several benchmark cases. We then develop a deep learning scheme for the reflected quadratic BSDE. The convergence analysis uses discrete reflection and truncation to reduce the quadratic problem to a globally Lipschitz system and combines reflected BSDE discretization estimates with neural network approximation errors. Numerical experiments for American options illustrate the effects of discount rate and entropic ambiguity on stopping values and exercise decisions.

1 Introduction

Dynamic risk measures have become a fundamental tool for the quantitative assessment of financial positions under uncertainty (see, among many others, [1, 9, 17, 7, 2, 4, 11]). Unlike classical conditional expectations, dynamic risk measures allow nonlinear preferences, model uncertainty, recursive evaluation and time-consistent risk assessment. Following the pioneering works on BSDEs and g -expectations [14, 24, 8, 5], it is now well understood that a large class of dynamic risk measures admits a representation through backward stochastic differential equations (BSDEs), where the generator completely characterizes the underlying notion of risk, ambiguity and investor preferences. This connection has subsequently been developed in the context of nonlinear pricing, coherent and convex risk measures, and utility valuation [15, 26, 2, 10].

Among the various classes of dynamic risk measures, cash-subadditive risk measures have attracted considerable attention because they naturally incorporate ambiguity in discounting and interest rates. Unlike the classical cash-additive framework, they allow uncertainty in the time value of money and therefore provide a more flexible approach to long term financial valuation. Their BSDE characterization and dual representation were established by El Karoui and Ravanelli [16], who showed that discount rate ambiguity induces a nonlinear dependence

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of the driver on the value process. See also, e.g., [12, 23] for a discussion on cash-subadditive risk measures.

Optimal stopping under dynamic risk measures extends the classical theory of optimal stopping by replacing conditional expectations with nonlinear dynamic risk evaluations. Such problems arise naturally in the valuation of American derivatives under model uncertainty, optimal liquidation, insurance contracts, real options, and investment timing problems. In the Brownian framework, Bayraktar, Karatzas and Yao [3] studied optimal stopping for convex dynamic risk measures, while Quenez and Sulem [25] established a reflected BSDE representation for robust optimal stopping under dynamic risk measures with jumps, thereby extending the nonlinear Snell envelope approach initiated by El Karoui et al. [13].

The objective of the present paper is to develop a unified framework for optimal stopping under *paired ambiguity*, where uncertainty simultaneously affects the probabilistic model and the discount mechanism. The first source of ambiguity is represented through changes of probability measure via Girsanov transformations, while the second corresponds to uncertainty in stochastic discounting, following the cash-subadditive framework of [16]. Combining these two mechanisms leads to a new family of dynamic risk measures (called *paired-ambiguity dynamic risk measures*) that naturally generalizes both classical entropic risk measures and cash-subadditive risk measures.

Our first contribution is to formulate the corresponding optimal stopping problem and to characterize its value process by means of an upper reflected BSDE. The upper obstacle naturally appears because the problem is formulated in terms of liabilities, so that immediate stopping corresponds to the negative of the payoff process. We establish a reflected BSDE representation theorem and show that the optimal stopping strategy is given by the first hitting time of the reflected solution against the obstacle.

Our second contribution concerns the structural properties of the resulting stopping operator. We prove that it inherits the principal axioms of dynamic risk measures and investigate its geometric properties, including monotonicity, cash-subadditivity, cash-additivity, quasi-concavity, concavity under affine drivers, and the failure of convexity in general. These results indicate that the stopping operator should be viewed as a dynamic risk measure acting on adapted stochastic processes rather than merely on terminal random variables.

Special attention is devoted to quadratic drivers associated with entropic risk measures. Quadratic BSDEs play a central role in mathematical finance through their connection with exponential utility, entropic risk measures and robust valuation [21, 2, 12]. We revisit this framework in the reflected setting and derive the corresponding representation of entropic stopping risk measures. In particular, we obtain explicit stopping rules in several tractable situations and illustrate how discount ambiguity and model ambiguity interact within the reflected BSDE framework. The existence and uniqueness of reflected quadratic BSDEs rely on the theory developed by Kobylanski et al. [22].

Finally, we propose a deep learning approximation scheme for solving the resulting reflected BSDEs. Since the drivers considered in this paper may exhibit quadratic growth in the control variable, standard convergence results for Lipschitz BSDEs are no longer directly applicable. Deep schemes for reflected BSDEs with Lipschitz drivers have been developed both by hard projection [19] and by regularization [18]; for quadratic drivers without reflection the truncation analysis of [6] is available, and the reflected quadratic case has been treated in [27] for a scheme in which the conditional expectations are computed exactly. We combine the last two, the truncation being made exact by the uniform bound on the integrand of the discretely reflected equation, and obtain a convergence estimate in which the network approximation errors appear explicitly. Numerical experiments for American options demonstrate the practical performance of the method and illustrate the impact of ambiguity on optimal stopping decisions.

The paper is organized as follows. Section 2 introduces the paired-ambiguity dynamic risk measure and its dual representation. Section 3 formulates the optimal stopping problem and

derives the reflected BSDE characterization together with the main properties of the stopping operator. Section 4 is devoted to the entropic case and explicit stopping representations. Section 5 develops the deep learning approximation scheme and establishes its convergence. Section 6 presents financially meaningful examples of dynamic risk measures generated by reflected BSDEs. Finally, Section 7 reports numerical experiments illustrating the theoretical results.

2 Paired-Ambiguity Risk Measures and Reflected BSDE Representation

In this section we formulate the optimal stopping problem associated with the paired-ambiguity dynamic risk measure introduced here below and derive its reflected BSDE representation. The framework combines two different sources of uncertainty. The first corresponds to ambiguity in the underlying probabilistic model and is represented through Girsanov changes of measure. The second corresponds to ambiguity in stochastic discounting, leading to a cash-subadditive dynamic evaluation. As a consequence, the resulting stopping problem extends the classical nonlinear optimal stopping framework to a setting where both model uncertainty and discount rate uncertainty are treated simultaneously.

The formulation is inspired by the optimal stopping problem for convex dynamic risk measures studied by Bayraktar, Karatzas and Yao [3], while incorporating the cash-subadditive dual representation introduced by El Karoui and Ravanelli [16]. The combination of these two ingredients leads naturally to an upper reflected BSDE, which characterizes both the stopping value and the optimal exercise strategy.

2.1 Paired-ambiguity optimal stopping problem

Consider a bounded adapted payoff process

$$\xi = (\xi_t)_{0 \leq t \leq T}.$$

For every stopping time $\tau \in \mathcal{T}_t$, the paired-ambiguity dynamic risk measure is defined by

$$\rho_{t,\tau}(\xi_\tau) = \sup_{(\alpha,\beta) \in \mathcal{A}} \mathbb{E}^{\mathbb{Q}^\alpha} \left[e^{-\int_t^\tau \beta_s ds} (-\xi_\tau) - \int_t^\tau e^{-\int_t^s \beta_u du} g^*(s, \alpha_s, \beta_s) ds \middle| \mathcal{F}_t \right], \quad (1)$$

where the process α models ambiguity in the probability measure through Girsanov transformations, β represents ambiguity in stochastic discounting, and $\mathcal{A}$ is a given set over which there is paired-ambiguity. The function g^* denotes the convex conjugate of the BSDE driver g .

The corresponding optimal stopping problem is

$$V_t = \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \rho_{t,\tau}(\xi_\tau). \quad (2)$$

Unlike the classical cash-additive framework, the evaluation (1) simultaneously accounts for uncertainty in the probabilistic model and in the discount mechanism. Consequently, the stopping value incorporates both model ambiguity and interest rate ambiguity within a unified dynamic risk measure framework. Note that (1) corresponds to the dual representation of El Karoui and Ravanelli [16] for $\mathcal{A}$ as in (7). Here, instead, $\mathcal{A}$ is considered general.

The representation of the value process follows the same general philosophy as nonlinear Snell envelopes generated by BSDEs. However, the presence of the discount process modifies the dual representation and changes the associated BSDE driver. The reflected BSDE obtained below therefore extends the standard representation by incorporating both sources of ambiguity into a single nonlinear evaluation.

2.2 Standing assumptions

We work on a complete filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, \mathbb{P})$ satisfying the usual conditions and carrying a d -dimensional Brownian motion W . For $t \in [0, T]$, let $\mathcal{T}_t$ denote the set of $(\mathcal{F}_s)$ -stopping times with values in $[t, T]$. We use the standard spaces

$$\mathcal{S}^2 = \left\{ Y : Y \text{ is adapted and continuous, } \mathbb{E} \left[\sup_{0 \leq t \leq T} |Y_t|^2 \right] < \infty \right\}$$

and

$$\mathcal{H}^2 = \left\{ Z : Z \text{ is progressively measurable and } \mathbb{E} \left[\int_0^T |Z_t|^2 dt \right] < \infty \right\}.$$

Let $\xi = (\xi_t)_{t \in [0, T]}$ be an adapted bounded continuous process with $\xi \in \mathcal{S}^2$. The process ξ represents the financial payoff received upon stopping, so the upper obstacle for the risk process is $-\xi$ and the terminal condition is compatible with the obstacle, namely $Y_T = -\xi_T$. Let

$$g : \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$$

be a progressively measurable driver. We assume the following:

- $g(t, \cdot, \cdot)$ is continuous with linear-quadratic growth, i.e. there exists a constant $C > 0$ such that for any $y \in \mathbb{R}$, $z \in \mathbb{R}^d$

$$|g(t, y, z)| \leq |g(t, 0, 0)| + C|y| + C|z|^2, \quad g(\cdot, 0, 0) \in \mathcal{H}^2.$$

- For any $M > 0$ there exists $C > 0$ such that for any $y \in [-M, M]$, $z \in \mathbb{R}^d$

$$\left| \frac{\partial g}{\partial z}(t, y, z) \right| \leq C(1 + |z|).$$

- For any $\epsilon > 0$ there exists $C_\epsilon > 0$ such that for any $y \in \mathbb{R}$, $z \in \mathbb{R}^d$

$$\frac{\partial g}{\partial y}(t, y, z) \leq C_\epsilon + \epsilon|z|^2.$$

Note that the last assumption is automatically satisfied if g is differentiable and decreasing in y .

The assumptions guarantee that the fixed horizon BSDEs and the upper reflected BSDE below have a unique solution and the Comparison Theorem holds for both the BSDEs and the reflected BSDEs. See Theorems 2.3 and 2.6 of Kobylanski [21] for BSDEs and Corollary 1 and Proposition 3.2 of Kobylanski et al. [22] for reflected BSDEs. Consider a BSDE with driver g of the form

$$Y_t = -\xi_T + \int_t^T g(s, Y_s, Z_s) ds - \int_t^T Z_s dW_s, \quad 0 \leq t \leq T, \quad (3)$$

We recall the following definition of (upper) reflected BSDEs. See [13, 22] and the references therein.

Definition 2.1 (Upper reflected BSDE). Let $L = (L_t)_{0 \leq t \leq T}$ be an adapted obstacle process and let η be an $\mathcal{F}_T$ -measurable terminal condition satisfying

$$\eta \leq L_T.$$

A triple (Y, Z, K) is called a solution of the upper reflected BSDE with generator g , terminal condition η , and upper obstacle L if

$$Y_t = \eta + \int_t^T g(s, Y_s, Z_s) ds - \int_t^T Z_s dW_s - (K_T - K_t), \quad 0 \leq t \leq T, \quad (4)$$

where K is an adapted nondecreasing process with $K_0 = 0$, such that

$$Y_t \leq L_t, \quad 0 \leq t \leq T, \quad (5)$$

and

$$\int_0^T (Y_t - L_t) dK_t = 0. \quad (6)$$

The last condition is the Skorokhod minimality condition; equivalently, K can increase only on the contact set

$$\{t \in [0, T] : Y_t = L_t\}.$$

3 Optimal stopping problem for risk measures and obstacle

Let g^* denote the convex conjugate associated with the dual variables corresponding to model and discount ambiguity. Following the dual representation of cash-subadditive risk measures [16], we consider admissible pairs (α, β) in the class

$$\mathcal{A} = \{(\alpha, \beta) : \alpha \in \text{BMO}, \quad 0 \leq \beta_t \leq C, \quad g^*(t, \alpha_t, \beta_t) < +\infty\}. \quad (7)$$

For each admissible α , the stochastic exponential

$$\mathcal{E} \left(- \int_0^\cdot \alpha_s dW_s \right)_T = \exp \left\{ - \int_0^T \alpha_s dW_s - \frac{1}{2} \int_0^T |\alpha_s|^2 ds \right\}$$

is assumed to be a uniformly integrable martingale and defines an equivalent probability measure $\mathbb{Q}^\alpha$ by

$$\frac{d\mathbb{Q}^\alpha}{d\mathbb{P}} = \mathcal{E} \left(- \int_0^\cdot \alpha_s dW_s \right)_T.$$

The process β represents ambiguity in the stochastic discount rate. We also assume that the discounted penalty term is integrable for every $(\alpha, \beta) \in \mathcal{A}$, and that $\mathcal{A}$ is stable under stopping and pasting: if two admissible pairs are used before and after a stopping time, then the pasted pair is again admissible. Finally, the dual representation below is assumed to be time consistent, equivalently generated by the BSDE driver in (3). These stability and time consistency conditions are the hypotheses needed for the nonlinear Snell envelope and reflected BSDE arguments.

For $t \leq \tau$, the paired-ambiguity risk evaluation is

$$\rho_{t,\tau}(\xi_\tau) = \sup_{(\alpha,\beta) \in \mathcal{A}} \mathbb{E}^{\mathbb{Q}^\alpha} \left[e^{-\int_t^\tau \beta_s ds} (-\xi_\tau) - \int_t^\tau e^{-\int_t^s \beta_u du} g^*(s, \alpha_s, \beta_s) ds \middle| \mathcal{F}_t \right], \quad (8)$$

where

$$g^*(s, \alpha, \beta) = \sup_{(y,z) \in \mathbb{R} \times \mathbb{R}^d} \{-\beta y - \alpha \cdot z - g(s, y, z)\}. \quad (9)$$

See Theorem 7.5 of [16].

Under the standing assumptions, we now focus on the following optimal stopping problem

$$\varphi_t(\xi) := \text{ess inf}_{\tau \in \mathcal{T}_t} \rho_{t,\tau}(\xi_\tau). \quad (10)$$

The following result is the upper obstacle, risk measure counterpart of Proposition 3.1 in [22]. After the corresponding sign transformation, the essential supremum characterization therein becomes the essential infimum characterization below.

Theorem 3.1 (Reflected BSDE representation). *Under the standing assumptions 2.2, the value process $(\varphi_t(\xi))_t$ in (10) is the first component of the unique upper reflected BSDE (4)–(6), with*

$$\eta = -\xi_T, \quad L_t = -\xi_t.$$

In particular,

$$\varphi_t(\xi) = Y_t.$$

Moreover, under the regularity assumptions ensuring attainment of the optimal stopping problem, the first contact time

$$\tau_t^* = \inf\{s \geq t : Y_s = -\xi_s\}$$

is optimal.

Note that the stopping payoff is received as a financial payoff ξ_t , while the risk process is written with the sign convention of a liability. Hence the immediate stopping risk is $-\xi_t$, and the reflected BSDE has the upper obstacle $(-\xi_t)$.

In the following result, we prove the properties satisfied by the optimal stopping value φ .

For $m_t \in L^\infty(\mathcal{F}_t)$, we use the notation $\xi + m_t$ for the shifted payoff process defined on $[t, T]$ by

$$(\xi + m_t)_s := \xi_s + m_t, \quad s \in [t, T].$$

Theorem 3.2 (Dynamic risk measure properties and geometry of the stopping operator). *The stopping operator φ defined in (10) inherits the main axioms of dynamic risk measure. More precisely, it satisfies the following properties.*

- a) *(decreasing monotonicity) For any $t \in [0, T]$, φ_t is monotone decreasing, i.e., if $\xi_s \geq \eta_s$ for any $s \in [0, T]$ (with η being a bounded adapted obstacle), then $\varphi_t(\xi) \leq \varphi_t(\eta)$.*
- b) *(cash-subadditivity) If g is decreasing in y , then φ_t is cash-subadditive for any $t \in [0, T]$, i.e., $\varphi_t(\xi + m_t) \geq \varphi_t(\xi) - m_t$ for any obstacle ξ and $m_t \in L_+^\infty(\mathcal{F}_t)$.*
- c) *(cash-additivity) If g is independent of y , then φ_t is cash-additive for any $t \in [0, T]$, i.e., $\varphi_t(\xi + m_t) = \varphi_t(\xi) - m_t$ for any obstacle ξ and $m_t \in L^\infty(\mathcal{F}_t)$.*
- d) *(quasi-concavity) Assume that any $\rho_{t,\tau}$ satisfies the following continuity property:*

$$\rho_{t,\tau}(\text{ess sup}\{X; Y\}) = \text{ess inf}\{\rho_{t,\tau}(X); \rho_{t,\tau}(Y)\}.$$

If $\varphi_t(\xi) \geq c_t$ and $\varphi_t(\eta) \geq c_t$ for bounded obstacles ξ, η and for some $t \in [0, T]$ and $c_t \in L^\infty(\mathcal{F}_t)$, then $\varphi_t(\lambda\xi + (1-\lambda)\eta) \geq c_t$ for any $\lambda \in [0, 1]$.

e) *If g is affine in y, z , i.e. $g(t, y, z) = ay + b \cdot z + c$ for some $a, c \in \mathbb{R}$ and $b \in \mathbb{R}^d$, then φ_t is concave for any $t \in [0, T]$, i.e., $\varphi_t(\lambda\xi + (1-\lambda)\eta) \geq \lambda\varphi_t(\xi) + (1-\lambda)\varphi_t(\eta)$ for all obstacles ξ, η and $\lambda \in [0, 1]$.*

f) *φ_t is not convex in general. For instance, convexity fails when $g(t, y, 0) = 0$ for any $y \in \mathbb{R}$ or when g is independent of y and $g(t, 0) = 0$.*

The previous result highlights that convexity of φ fails in general and that the sufficient conditions (e.g., the continuity assumption in d) and the assumption on g in e)) for φ to be (quasi-)concave are quite strong.

Proof. a) Let ξ, η be adapted and bounded obstacles with $\xi_s \geq \eta_s$ for any $s \in [0, T]$.

By the properties of BSDEs and of the associated risk measures (see, e.g., [2], [14], [26]), $\rho_{t,\tau}^g(\xi_\tau) \leq \rho_{t,\tau}^g(\eta_\tau)$ for any $t \leq \tau$. It then follows that $\varphi_t(\xi) \leq \varphi_t(\eta)$ for all $t \in [0, T]$.

b) By El Karoui and Ravanelli [16], decreasing monotonicity of g in y implies cash-subadditivity of any $\rho_{t,\tau}$. It then follows that for any obstacle ξ and any $m_t \in L_+^\infty(\mathcal{F}_t)$

$$\begin{aligned}\varphi_t(\xi + m_t) &= \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \rho_{t,\tau}(\xi + m_t) \\ &\geq \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \{\rho_{t,\tau}(\xi) - m_t\} \\ &= \varphi_t(\xi) - m_t.\end{aligned}$$

c) By Briand et al. [5], independence of g in y implies cash-additivity of any $\rho_{t,\tau}$. The thesis then follows by proceeding as in item b).

d) Clearly, $\varphi_t(\xi) \geq c_t$ implies that $\rho_{t,\tau}(\xi_\tau) \geq c_t$ for any $\tau \in \mathcal{T}_t$; similarly for $\varphi_t(\eta)$.

For any $\lambda \in [0, 1]$

$$\begin{aligned}\varphi_t(\lambda\xi + (1-\lambda)\eta) &= \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \rho_{t,\tau}(\lambda\xi_\tau + (1-\lambda)\eta_\tau) \\ &\geq \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \rho_{t,\tau}(\operatorname{ess\,sup}\{\xi_\tau; \eta_\tau\}) \\ &= \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \operatorname{ess\,inf}\{\rho_{t,\tau}(\xi_\tau); \rho_{t,\tau}(\eta_\tau)\} \\ &\geq c_t,\end{aligned}\tag{11}$$

where (11) is due to the continuity assumption of $\rho_{t,\tau}$.

e) If g is affine, then the associated $\rho_{t,s}$ is both convex and concave (see, e.g., [2], [14], [26]). It then follows that, for any $\lambda \in [0, 1]$ and all obstacles ξ, η ,

$$\begin{aligned}\varphi_t(\lambda\xi + (1-\lambda)\eta) &= \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \rho_{t,\tau}(\lambda\xi_\tau + (1-\lambda)\eta_\tau) \\ &\geq \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \{\lambda\rho_{t,\tau}(\xi_\tau) + (1-\lambda)\rho_{t,\tau}(\eta_\tau)\} \\ &\geq \lambda \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \rho_{t,\tau}(\xi_\tau) + (1-\lambda) \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \rho_{t,\tau}(\eta_\tau) \\ &= \lambda\varphi_t(\xi) + (1-\lambda)\varphi_t(\eta).\end{aligned}$$

f) Assume that $g(t, y, 0) = 0$ for any $y \in \mathbb{R}$ or that g is independent of y and $g(t, 0) = 0$. Then, by [24] (see also Prop. 2.1 of [8]), the associated $\rho_{t,s}$ satisfies constancy, i.e. $\rho_{t,s}(c_t) = -c_t$ for any $c_t \in L^\infty(\mathcal{F}_t)$.

For deterministic obstacles ξ, η , then,

$$\varphi_t(\xi) = \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \rho_{t,\tau}(\xi_\tau) = \operatorname{ess\,inf}_{s \in [t, T]} \rho_{t,s}(\xi_s) = \operatorname{ess\,inf}_{s \in [t, T]} (-\xi_s),$$

where the last equality is due to the constancy property. A similar expression holds for $\varphi_t(\eta)$.

Let $t \in [0, T)$ and take $\xi_s = s$, $\eta_s = -2s$ for any $s \in [t, T]$ and $\lambda = \frac{1}{2}$. It then follows immediately that

$$\varphi_t(\xi) = \operatorname{ess\,inf}_{s \in [t, T]} (-\xi_s) = -T; \quad \varphi_t(\eta) = \operatorname{ess\,inf}_{s \in [t, T]} (-\eta_s) = 2t$$

and

$$\varphi_t(\lambda\xi + (1-\lambda)\eta) = \operatorname{ess\,inf}_{s \in [t, T]} \left(-\frac{1}{2}\xi_s - \frac{1}{2}\eta_s \right) = \frac{t}{2} > \frac{1}{2}\varphi_t(\xi) + \frac{1}{2}\varphi_t(\eta) = -\frac{T}{2} + t,$$

where the strict inequality holds for any $t < T$. Convexity then fails in general. $\square$

Note that the previous proof is based on the relation between optimal stopping problems, reflected BSDEs, and BSDEs.

It is worth underlining that, by its definition and by its properties, φ can be seen more as a dynamic risk measure for processes than as one for random variables.

In the following, we will focus our attention on examples of drivers of the following form:

$$g(t, y, z) = \sup_{(\delta, \gamma) \in \mathcal{C}} \psi(\delta, \gamma, y, z),$$

for a given set $\mathcal{C}$ that can be interpreted as the set over which there is paired-ambiguity and for $\psi(\delta, \gamma, y, z)$ convex in (y, z) and decreasing in y . Consequently, the associated driver g satisfies the same properties.

Some special cases of drivers could be:

- $g(t, y, z) = \sup_{(\delta, \gamma) \in \mathcal{C}} \{-\delta y + \frac{\gamma}{2}|z|^2\}$ with $\delta > 0$ on $\mathcal{C}$ that can be seen as a paired-ambiguity on aversion in the entropic-type term and on discounting;
- $g(t, y, z) = \sup_{\delta \in \mathcal{C}} \{-\delta y\}$ with $\delta > 0$ on $\mathcal{C}$ (ambiguity on discounting);
- $g(t, y, z) = \sup_{\gamma \in \mathcal{C}} \{\gamma z\}$ (ambiguity on the probability measure/generalized scenario Q over which expectation is taken);
- $g(t, y, z) = \sup_{(\delta, \gamma) \in \mathcal{C}} \{-\delta y + \gamma z\}$ with $\delta > 0$ on $\mathcal{C}$ (linear pair-ambiguity);
- $g(t, y, z) = \sup_{\delta} \{-\delta y + \bar{\gamma} z\}$ with $\delta > 0$ on $\mathcal{C}$;
- $g(t, y, z) = \sup_{\gamma} \{\frac{\gamma}{2}|z|^2\}$ (ambiguity on the aversion parameter in the entropic case).

4 Entropic Risk Measure

The entropic case is the canonical reflected quadratic BSDE example for this stopping problem. It should be treated separately from the Lipschitz theory: existence, uniqueness and comparison rely on quadratic BSDE estimates, BMO martingales and exponential integrability conditions as in Kobylanski type results [21, 22]. In the reflected case, these conditions are combined with the upper obstacle estimates used in the Quenez–Sulem representation [25].

4.1 Properties of the entropic reflected quadratic BSDE

In the stopping problem, the relevant object is not the fixed terminal entropic BSDE alone, but the upper reflected quadratic BSDE with obstacle $-\xi$:

$$Y_t = -\xi_T + \int_t^T \frac{\gamma}{2} |Z_s|^2 ds - \int_t^T Z_s dW_s - (K_T - K_t), \quad 0 \leq t \leq T, \quad (12)$$

with

$$Y_t \leq -\xi_t, \quad \int_0^T (Y_t + \xi_t) dK_t = 0. \quad (13)$$

Here K is the reflection process. The ordinary entropic BSDE appears only as the local equation on an interval where no reflection occurs, or as the fixed stopping time evaluation used inside the nonlinear Snell envelope.

4.2 Explicit entropic representation and reflected envelope

We now exploit the exponential structure of the entropic driver to obtain a more explicit representation of the stopping problem. The exponential transformation converts the upper reflected quadratic BSDE into a classical lower Snell envelope problem for the transformed payoff $e^{-\gamma\xi}$. This yields both an explicit representation of the value process and a direct characterization of the associated optimal stopping rule.

Proposition 4.1 (Exponential transform for the entropic reflected quadratic BSDE). *Let $\gamma > 0$, and assume that the payoff process ξ is continuous and has sufficient exponential moments so that the quadratic reflected BSDE below is well posed in the sense of the quadratic reflected BSDE literature. Let $g(z) = (\gamma/2)|z|^2$, and denote by $\mathcal{E}_{t,\tau}^g[\eta]$ the first component at time t of the BSDE on $[t, \tau]$ with driver g and terminal condition η . In the entropic case,*

$$\mathcal{E}_{t,\tau}^g[-\xi_\tau] = \frac{1}{\gamma} \log \mathbb{E} \left[e^{-\gamma \xi_\tau} \mid \mathcal{F}_t \right]. \quad (14)$$

Then the value process

$$V_t := \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \mathcal{E}_{t,\tau}^g[-\xi_\tau] \quad (15)$$

is characterized by the upper reflected quadratic BSDE

$$V_t = -\xi_T + \int_t^T \frac{\gamma}{2} |Z_s|^2 ds - \int_t^T Z_s dW_s - (K_T - K_t), \quad (16)$$

with obstacle and Skorokhod conditions

$$V_t \leq -\xi_t, \quad \int_0^T (V_t + \xi_t) dK_t = 0. \quad (17)$$

Moreover, when the obstacle regularity ensures optimality of the first hitting time, an optimal stopping time is

$$\tau_t^* = \inf\{s \geq t : V_s = -\xi_s\}.$$

Equivalently, this BSDE nonlinear expectation value has the explicit reflected envelope representation

$$V_t = \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \mathcal{E}_{t,\tau}^g[-\xi_\tau] = \frac{1}{\gamma} \log \left(\operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \mathbb{E} \left[e^{-\gamma \xi_\tau} \mid \mathcal{F}_t \right] \right). \quad (18)$$

Proof. We first recall the non-reflected entropic calculation, since it is the basic exponential transform behind the reflected formula. Fix a stopping time $\tau \in \mathcal{T}_t$ and a terminal payoff ξ_τ with the required exponential integrability. The quadratic BSDE on $[t, \tau]$ with terminal value $-\xi_\tau$ is

$$Y_s^\tau = -\xi_\tau + \int_s^\tau \frac{\gamma}{2} |Z_r^\tau|^2 dr - \int_s^\tau Z_r^\tau dW_r, \quad t \leq s \leq \tau. \quad (19)$$

Set $M_s^\tau := \exp(\gamma Y_s^\tau)$. By Itô's formula,

$$\begin{aligned} dM_s^\tau &= \gamma M_s^\tau dY_s^\tau + \frac{\gamma^2}{2} M_s^\tau |Z_s^\tau|^2 ds \\ &= \gamma M_s^\tau \left(-\frac{\gamma}{2} |Z_s^\tau|^2 ds + Z_s^\tau dW_s \right) + \frac{\gamma^2}{2} M_s^\tau |Z_s^\tau|^2 ds \\ &= \gamma M_s^\tau Z_s^\tau dW_s. \end{aligned}$$

Thus M^τ is a martingale under the stated exponential moment assumptions, and its terminal value is $M_\tau^\tau = e^{-\gamma \xi_\tau}$. Consequently

$$Y_t^\tau = \frac{1}{\gamma} \log \mathbb{E} \left[e^{-\gamma \xi_\tau} \mid \mathcal{F}_t \right] = \mathcal{E}_{t,\tau}^g[-\xi_\tau]. \quad (20)$$

Taking the essential infimum over all admissible stopping times gives

$$V_t = \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} Y_t^\tau = \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \mathcal{E}_{t,\tau}^g[-\xi_\tau] = \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \frac{1}{\gamma} \log \mathbb{E} \left[e^{-\gamma \xi_\tau} \mid \mathcal{F}_t \right].$$

Since $x \mapsto (1/\gamma) \log x$ is increasing, this is equivalent to (18). If

$$R_t := \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \mathbb{E} \left[e^{-\gamma \xi_\tau} \mid \mathcal{F}_t \right],$$

then R is the lower Snell envelope of the transformed payoff $e^{-\gamma \xi}$. In particular, $R_t \leq e^{-\gamma \xi_t}$, and applying the increasing map $(1/\gamma) \log(\cdot)$ gives the upper obstacle condition $V_t \leq -\xi_t$.

We now identify explicitly how the reflection term transforms under the exponential change of variables. Since

$$R_t = e^{\gamma V_t},$$

and the reflected quadratic BSDE can be written in differential form as

$$dV_t = -\frac{\gamma}{2} |Z_t|^2 dt + Z_t dW_t + dK_t,$$

Itô's formula gives

$$\begin{aligned} dR_t &= \gamma R_t dV_t + \frac{\gamma^2}{2} R_t |Z_t|^2 dt \\ &= \gamma R_t \left(-\frac{\gamma}{2} |Z_t|^2 dt + Z_t dW_t + dK_t \right) + \frac{\gamma^2}{2} R_t |Z_t|^2 dt \\ &= \gamma R_t Z_t dW_t + \gamma R_t dK_t. \end{aligned}$$

Thus, setting

$$\tilde{Z}_t := \gamma R_t Z_t$$

and

$$A_t := \int_0^t \gamma R_s dK_s,$$

we obtain

$$dR_t = \tilde{Z}_t dW_t + dA_t.$$

Since K is nondecreasing and $R_t > 0$, the process A is also nondecreasing.

Moreover, the upper obstacle condition $V_t \leq -\xi_t$ is equivalent, under the increasing exponential transformation, to

$$R_t \leq e^{-\gamma \xi_t}.$$

The Skorokhod condition is preserved by the transformation. Indeed, dK_t is carried by the contact set

$$\{V_t = -\xi_t\},$$

and therefore $dA_t = \gamma R_t dK_t$ is carried by

$$\{R_t = e^{-\gamma \xi_t}\}.$$

Consequently,

$$\int_0^T (R_t - e^{-\gamma \xi_t}) dA_t = 0.$$

Hence R is precisely the lower Snell envelope of the transformed payoff $e^{-\gamma \xi}$, with Doob–Meyer decomposition

$$dR_t = \tilde{Z}_t dW_t + dA_t.$$

Conversely, starting from the lower Snell envelope R , define

$$V_t := \frac{1}{\gamma} \log R_t, \quad Z_t := \frac{\tilde{Z}_t}{\gamma R_t}, \quad K_t := \int_0^t \frac{1}{\gamma R_s} dA_s.$$

Applying Itô's formula to $V_t = (1/\gamma) \log R_t$ yields

$$dV_t = -\frac{\gamma}{2}|Z_t|^2 dt + Z_t dW_t + dK_t.$$

Together with

$$V_t \leq -\xi_t, \quad \int_0^T (V_t + \xi_t) dK_t = 0,$$

this recovers the upper reflected quadratic BSDE

$$V_t = -\xi_T + \int_t^T \frac{\gamma}{2}|Z_s|^2 ds - \int_t^T Z_s dW_s - (K_T - K_t).$$

Conversely, the comparison and minimality principle for reflected BSDEs identify the first component of any solution with the essential infimum of the entropic stopping evaluations. The first contact time $\tau_t^* = \inf\{s \geq t : V_s = -\xi_s\}$ is optimal whenever the standard obstacle regularity assumptions ensure that this hitting time is admissible and the Snell envelope is attained there. $\square$

The proposition above contains the detailed exponential transform proof. The key point is that reflection does not destroy the entropic transform; it changes the single conditional expectation in the non-reflected formula (20) into the lower Snell envelope appearing in (18). Thus the remaining object is classical: the optimal stopping envelope of the transformed payoff $e^{-\gamma\xi}$.

The proposition above reduces the entropic stopping problem to a classical optimal stopping problem for the positive transformed payoff

$$U_t := e^{-\gamma\xi_t}.$$

In general, the lower Snell envelope of U is the explicit object; a closed formula requires additional structure on ξ . The following cases are useful benchmarks for checking a numerical solver.

Proposition 4.2 (Explicit stopping solutions for entropic risk). *Let*

$$U_t := e^{-\gamma\xi_t}, \quad R_t := \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \mathbb{E}[U_\tau \mid \mathcal{F}_t], \quad V_t := \frac{1}{\gamma} \log R_t.$$

Then the following cases admit explicit optimal stopping rules.

(i) *If U is a submartingale, then*

$$R_t = U_t = e^{-\gamma\xi_t}, \quad V_t = -\xi_t,$$

and immediate stopping

$$\tau_t^* = t$$

is optimal.

(ii) *If U is a supermartingale, then*

$$R_t = \mathbb{E}[U_T \mid \mathcal{F}_t] = \mathbb{E}[e^{-\gamma\xi_T} \mid \mathcal{F}_t],$$

and therefore

$$V_t = \frac{1}{\gamma} \log \mathbb{E}[e^{-\gamma\xi_T} \mid \mathcal{F}_t].$$

In this case terminal stopping

$$\tau_t^* = T$$

is optimal.

(iii) If $\xi = (\xi_s)_{0 \leq s \leq T}$ is deterministic and continuous, then

$$R_t = \min_{s \in [t, T]} e^{-\gamma \xi_s} = e^{-\gamma \max_{s \in [t, T]} \xi_s},$$

and hence

$$V_t = - \max_{s \in [t, T]} \xi_s.$$

An optimal stopping time is the first future maximizer

$$\tau_t^* = \inf \left\{ s \in [t, T] : \xi_s = \max_{u \in [t, T]} \xi_u \right\}.$$

(iv) Suppose that

$$\xi_t = x_0 + \mu t + \sigma W_t, \quad \sigma \neq 0.$$

Then

$$U_t = e^{-\gamma x_0} \exp(-\gamma \mu t - \gamma \sigma W_t),$$

and

$$dU_t = U_t \left[\left(-\gamma \mu + \frac{\gamma^2 \sigma^2}{2} \right) dt - \gamma \sigma dW_t \right].$$

Consequently, if

$$\mu < \frac{\gamma \sigma^2}{2},$$

then U is a submartingale, immediate stopping is optimal, and

$$R_t = e^{-\gamma \xi_t}, \quad V_t = -\xi_t.$$

If

$$\mu > \frac{\gamma \sigma^2}{2},$$

then U is a supermartingale, terminal stopping is optimal, and

$$R_t = \exp \left(-\gamma \xi_t + \left[-\gamma \mu + \frac{\gamma^2 \sigma^2}{2} \right] (T - t) \right),$$

so that

$$V_t = -\xi_t + \left(-\mu + \frac{\gamma \sigma^2}{2} \right) (T - t).$$

At the critical value

$$\mu = \frac{\gamma \sigma^2}{2},$$

the process U is a martingale. Hence every stopping time $\tau \in \mathcal{T}_t$ gives the same value,

$$R_t = U_t, \quad V_t = -\xi_t.$$

Proof. For (i), if U is a submartingale, then for every $\tau \in \mathcal{T}_t$, the optional sampling theorem gives

$$\mathbb{E}[U_\tau \mid \mathcal{F}_t] \geq U_t.$$

Since $\tau = t$ is admissible,

$$R_t = \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \mathbb{E}[U_\tau \mid \mathcal{F}_t] = U_t.$$

Therefore

$$V_t = \frac{1}{\gamma} \log U_t = -\xi_t,$$

and immediate stopping is optimal.

For (ii), if U is a supermartingale, then for every $\tau \in \mathcal{T}_t$,

$$\mathbb{E}[U_T \mid \mathcal{F}_\tau] \leq U_\tau.$$

Taking conditional expectations with respect to $\mathcal{F}_t$ yields

$$\mathbb{E}[U_T \mid \mathcal{F}_t] \leq \mathbb{E}[U_\tau \mid \mathcal{F}_t].$$

Since $\tau = T$ is admissible,

$$R_t = \mathbb{E}[U_T \mid \mathcal{F}_t],$$

which gives the stated formula for V_t .

For (iii), since ξ is deterministic,

$$R_t = \min_{s \in [t, T]} e^{-\gamma \xi_s}.$$

Because $x \mapsto e^{-\gamma x}$ is decreasing,

$$\min_{s \in [t, T]} e^{-\gamma \xi_s} = e^{-\gamma \max_{s \in [t, T]} \xi_s}.$$

Continuity of ξ ensures that the maximum is attained, and therefore the first maximizer is an optimal stopping time.

Finally, for (iv), Itô's formula applied to $U_t = e^{-\gamma \xi_t}$ gives

$$dU_t = U_t \left[\left(-\gamma \mu + \frac{\gamma^2 \sigma^2}{2} \right) dt - \gamma \sigma dW_t \right].$$

Thus U is a submartingale when

$$\mu < \frac{\gamma \sigma^2}{2},$$

a supermartingale when

$$\mu > \frac{\gamma \sigma^2}{2},$$

and a martingale at equality. The corresponding optimal stopping rules follow from (i) and (ii).

In the supermartingale case, conditioning on the Brownian increment $W_T - W_t$ gives

$$\mathbb{E}[e^{-\gamma \xi_T} \mid \mathcal{F}_t] = \exp \left(-\gamma \xi_t + \left[-\gamma \mu + \frac{\gamma^2 \sigma^2}{2} \right] (T - t) \right),$$

and applying $(1/\gamma) \log(\cdot)$ yields

$$V_t = -\xi_t + \left(-\mu + \frac{\gamma \sigma^2}{2} \right) (T - t).$$

□

Remark 4.3 (Two Brownian benchmark examples). Proposition 4.2(iv) yields two particularly simple benchmark examples.

First, consider

$$\xi_t = aW_t - \frac{\gamma a^2}{2} t + ct, \quad c > 0.$$

Here

$$\mu = c - \frac{\gamma a^2}{2}, \quad \sigma = a.$$

Hence $U_t = e^{-\gamma \xi_t}$ is a supermartingale precisely when

$$c \geq \gamma a^2.$$

If $c > \gamma a^2$, terminal stopping is optimal and

$$V_t = -aW_t + \gamma a^2 T - \frac{\gamma a^2}{2}t - cT.$$

At the critical value $c = \gamma a^2$, U is a martingale, so every stopping time in $\mathcal{T}_t$ is optimal and

$$V_t = -\xi_t.$$

Second, consider

$$\xi_t = aW_t - \frac{\gamma a^2}{2}t - ct, \quad c > 0.$$

In this case

$$\mu = -c - \frac{\gamma a^2}{2} < \frac{\gamma a^2}{2},$$

so U is a submartingale. Consequently, immediate stopping is optimal,

$$\tau_t^* = t,$$

and

$$V_t = -\xi_t.$$

5 Algorithm and its Convergence

In this section we establish convergence of a deep learning scheme for the reflected BSDE of Section 2.2. Two obstructions have to be removed. The driver is quadratic in z , whereas the available error estimates for backward schemes require a globally Lipschitz driver; and the martingale integrand Z of a continuously reflected quadratic BSDE need not be bounded, so the driver cannot simply be truncated without changing the solution. Both are handled by passing through the discretely reflected equation, following [27], on which the integrand *is* uniformly bounded and the truncation is therefore exact. The resulting Lipschitz system is then treated by the reflected deep backward scheme of [19].

5.1 Markovian setting

Throughout this section we work in a Markovian framework, which is more restrictive than that of Section 2.2. Let X solve

$$X_t = x_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s) dW_s, \quad t \in [0, T], \quad (21)$$

let the obstacle be $\xi_t = h(X_t)$ for a deterministic h , and let the driver be $g(t, x, y, z)$. The upper reflected BSDE (4) then reads

$$\begin{cases} Y_t = -h(X_T) + \int_t^T g(s, X_s, Y_s, Z_s) ds - \int_t^T Z_s dW_s - (K_T - K_t), \\ Y_t \leq -h(X_t), \quad \int_0^T (Y_t + h(X_t)) dK_t = 0, \\ K \text{ continuous nondecreasing,} \quad K_0 = 0. \end{cases} \quad (22)$$

(H1) *Forward process.* b and σ are deterministic, σ does not depend on x , and for some $L > 0$

$$|b(t, 0)| + |\sigma(t)| \leq L, \quad |b(t, x) - b(t, x')| \leq L|x - x'|.$$

(H2) *Obstacle.* $h : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies $|h(x)| \leq M_h$ and $|h(x) - h(x')| \leq L|x - x'|$. No sign condition is imposed.

(H3) *Driver.* $g : [0, T] \times \mathbb{R}^d \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$ is continuous and, for constants $M_g, \gamma, L > 0$,

$$|g(t, x, y, z)| \leq M_g(1 + |y|) + \frac{\gamma}{2}|z|^2, \quad (23)$$

$$|g(t, x, y, z) - g(t, x', y, z)| \leq L(1 + |z|)|x - x'|, \quad (24)$$

$$|g(t, x, y, z) - g(t, x, y', z)| \leq L|y - y'|, \quad (25)$$

$$|g(t, x, y, z) - g(t, x, y, z')| \leq L(1 + |z| + |z'|)|z - z'|. \quad (26)$$

(H4) *Time regularity.* For $s \leq t$,

$$|b(t, x) - b(s, x)| + |\sigma(t) - \sigma(s)| + |g(t, x, y, z) - g(s, x, y, z)| \leq L(t - s)^{1/2}.$$

Two further conditions are used only to quantify the discretisation error:

(R1) $h \in C_b^1$ with L -Lipschitz derivative.

(R2) $h \in C_b^2$ with L -Lipschitz first and second derivatives, and σ is L -Lipschitz in t .

Remark 5.1. Relative to Section 2.2, (26) is the integrated form of the hypothesis $|\partial_z g| \leq C(1 + |z|)$ imposed there and requires no differentiability; (23) strengthens the linear-quadratic growth bound by requiring the y -independent part to be bounded rather than merely in $\mathcal{H}^2$. The genuine restriction is Markovianity, which is unavoidable: the networks below take X_{t_i} as input, so the scheme can only represent $Y_t = u(t, X_t)$. That the drivers of Section 6 satisfy (H3)–(H4) is verified in Remark 7.1.

It is convenient to record the reversal

$$\widehat{Y} := -Y, \quad \widehat{Z} := -Z, \quad \widehat{K} := K, \quad f(t, x, y, z) := -g(t, x, -y, -z), \quad (27)$$

under which (22) becomes the lower reflected equation

$$\widehat{Y}_t = h(X_T) + \int_t^T f(s, X_s, \widehat{Y}_s, \widehat{Z}_s) ds - \int_t^T \widehat{Z}_s dW_s + (\widehat{K}_T - \widehat{K}_t), \quad \widehat{Y}_t \geq h(X_t),$$

with $\widehat{K}$ nondecreasing. Since $|f(t, x, y, z)| = |g(t, x, -y, -z)|$ and the moduli in (H3) are invariant under $(y, z) \mapsto (-y, -z)$, f satisfies (H3)–(H4) with the same constants. The results of [27] and [19] are stated for the lower reflected equation and are quoted below for f ; all statements transfer to (22) through (27).

5.2 Discrete reflection and a priori bounds

Given a grid $\pi = \{0 = t_0 < \dots < t_N = T\}$ with $|\pi| = \max_i(t_{i+1} - t_i)$ and $N|\pi| \leq L$, let X^π denote the Euler scheme for (21), so that

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t - X_t^\pi|^{2p} \right] \leq C|\pi|^p, \quad p \geq 1, \quad (28)$$

which is the classical strong convergence estimate for the Euler–Maruyama scheme under (H1); see for instance [20]. For a set of reflection dates $R \subseteq \pi$ let (Y^R, Z^R) denote the solution of the discretely reflected equation, in which reflection acts only at the dates of R and the equation is solved exactly in between, and let $(Y^{R,e}, Z^{R,e})$ denote the same object driven by X^π rather than X . We write (Y^e, Z^e, K^e) for the solution of (22) driven by X^π .

Lemma 5.2 ([27], Prop. 2.2 and Lem. 4.5). *Under (H1)–(H3) there are constants M and M_z , depending only on L, M_g, M_h, γ and T , such that*

$$\|Y\|_\infty \leq M, \quad \|Z \cdot W\|_{\text{BMO}}^2 \leq \frac{e^{4\gamma M}}{\gamma^2} [1 + 2\gamma M_g(1 + M)T], \quad (29)$$

and, for every $R \subseteq \pi$,

$$|Z_t^R| \leq M_z, \quad |Z_t^{R,e}| \leq M_z, \quad 0 \leq t \leq T. \quad (30)$$

Remark 5.3. The bound (30) holds for the *discretely* reflected integrand, uniformly in R , and not for Z itself: the integrand of a continuously reflected quadratic BSDE need not be bounded, and (29) gives only a BMO estimate. This is the reason for introducing (Y^R, Z^R) at all. The hypothesis that σ be deterministic in (H1) is what secures (30); the case of state-dependent σ is open, see [27].

5.3 Truncated driver and the scheme

Let $\Pi_R(x) = \max(-R, \min(x, R))$ act componentwise and set

$$g^{M_z}(t, x, y, z) := g(t, x, y, \Pi_{M_z}(z)), \quad (31)$$

the truncation radius being fixed to the constant M_z of Lemma 5.2. Since $|\Pi_{M_z}(z)| \leq \sqrt{d} M_z$, (24)–(26) give, for all (x, y, z) and (x', y', z') ,

$$|g^{M_z}(t, x, y, z) - g^{M_z}(t, x', y', z')| \leq L(1 + \sqrt{d} M_z)|x - x'| + L|y - y'| + L(1 + 2\sqrt{d} M_z)|z - z'|, \quad (32)$$

so g^{M_z} is globally Lipschitz. By (30), $\Pi_{M_z}(Z_t^R) = Z_t^R$ and $\Pi_{M_z}(Z_t^{R,e}) = Z_t^{R,e}$, whence

$$(Y^{R, M_z}, Z^{R, M_z}) = (Y^R, Z^R), \quad (Y^{R, e, M_z}, Z^{R, e, M_z}) = (Y^{R, e}, Z^{R, e}) : \quad (33)$$

truncation changes nothing at the discretely reflected level.

We now define the scheme, taking $R = \pi$. At each date two networks $\mathcal{U}_i(\cdot; \theta)$ and $\mathcal{Z}_i(\cdot; \theta)$ approximate the value and the integrand.

Algorithm 1 projects at every date, so it is discretely reflected with $R = \pi$, the case of (33). The estimates of [27] below hold for general $R \subseteq \pi$; taking $R = \pi$ replaces their $|R|$ by $|\pi|$ and their $\alpha_i(\kappa)$ by $\alpha_i(N)$.

5.4 Convergence

Write $(\bar{Y}^\pi, \bar{Z}^\pi)$ for the discrete-time scheme obtained from Algorithm 1 when the conditional expectations are computed exactly, and let

$$\epsilon_i^{\mathcal{N}, \tilde{u}} := \inf_{\xi} \mathbb{E} |\tilde{u}_i(X_{t_i}^\pi) - \mathcal{U}_i(X_{t_i}^\pi; \xi)|^2, \quad \epsilon_i^{\mathcal{N}, \tilde{z}} := \inf_{\eta} \mathbb{E} |\tilde{z}_i(X_{t_i}^\pi) - \mathcal{Z}_i(X_{t_i}^\pi; \eta)|^2$$

be the L^2 approximation errors of the network classes for the conditional expectations $\tilde{u}_i, \tilde{z}_i$ arising at step i . The total error is measured by

$$\mathcal{E}[(\hat{\mathcal{U}}, \hat{\mathcal{Z}}), (Y, Z)] := \max_{i=0, \dots, N-1} \mathbb{E} |Y_{t_i} - \hat{\mathcal{U}}_i(X_{t_i}^\pi)|^2 + \mathbb{E} \left[\sum_{i=0}^{N-1} \int_{t_i}^{t_{i+1}} |Z_t - \hat{\mathcal{Z}}_i(X_{t_i}^\pi)|^2 dt \right]. \quad (36)$$

Proposition 5.4. *Let (H1)–(H4) hold and let $R = \pi$. There is a constant $C > 0$, independent of π and of the network classes, such that*

$$\mathcal{E}[(\hat{\mathcal{U}}, \hat{\mathcal{Z}}), (Y, Z)] \leq C \left(\varepsilon(\pi) + \sum_{i=0}^{N-1} (N \epsilon_i^{\mathcal{N}, \tilde{u}} + \epsilon_i^{\mathcal{N}, \tilde{z}}) \right), \quad (37)$$

Algorithm 1 Reflected deep backward dynamic programming with upper obstacle

Step 1. **Input.** Simulate the Euler paths $(X_{t_i}^\pi, \Delta W_i)_{i=0}^{N-1}$; the obstacle function h ; the truncated driver g^{M_z} of (31); network classes $\mathcal{U}_i, \mathcal{Z}_i$.

Step 2. **Terminal condition.** Set $\widehat{\mathcal{U}}_N(X_{t_N}^\pi) = -h(X_{t_N}^\pi)$.

Step 3. **Backward regression.** For $i = N - 1, \dots, 0$, given $\widehat{\mathcal{U}}_{i+1}$, minimise

$$\begin{aligned} \widehat{L}_i(\theta) := \mathbb{E} \Big[& \widehat{\mathcal{U}}_{i+1}(X_{t_{i+1}}^\pi) - \mathcal{U}_i(X_{t_i}^\pi; \theta) \\ & + g^{M_z}(t_i, X_{t_i}^\pi, \mathcal{U}_i(X_{t_i}^\pi; \theta), \mathcal{Z}_i(X_{t_i}^\pi; \theta)) \Delta t_i - \mathcal{Z}_i(X_{t_i}^\pi; \theta) \Delta W_i \Big|^2 \Big], \end{aligned} \quad (34)$$

and let θ_i^* be a minimiser.

Step 4. **Upper reflection.** Set

$$\widehat{\mathcal{U}}_i(x) = \min\{\mathcal{U}_i(x; \theta_i^*), -h(x)\}, \quad \widehat{\mathcal{Z}}_i(x) = \mathcal{Z}_i(x; \theta_i^*). \quad (35)$$

Step 5. **Output.** $(\widehat{\mathcal{U}}_i, \widehat{\mathcal{Z}}_i)_{i=0}^{N-1}$ and $\widehat{Y}_0 = \widehat{\mathcal{U}}_0(x_0)$.

where $\varepsilon(\pi)$ is the discretisation error. Under (R1), $\varepsilon(\pi) = O(|\pi|^{1/2})$ for the Y -component and $O(|\pi|^{1/2})$ for the Z -component of (36); under (R2) the Y -component improves to $O(|\pi|)$, while the Z -component does not improve.

Here these rates refer to the squared error quantities in (36). Equivalently, at the norm level, the corresponding rate for the Y -component is $O(|\pi|^{1/4})$ under (R1) and $O(|\pi|^{1/2})$ under (R2), while the H^2 error for the Z -component is $O(|\pi|^{1/4})$ under both assumptions.

Proof. Throughout, constants depend only on the data of (H1)–(H4) and on M_z . By (33) the driver may be replaced by g^{M_z} in every discretely reflected object without altering it, and by (32) that driver is globally Lipschitz, so all estimates below are for the Lipschitz system. The estimates for the passage to discrete reflection and for the time discretisation are those of [27], and the estimate for the network error is that of [19]; both are restated here for completeness and are not original to this paper. What is established below is their combination.

For the value component, introduce the discrete $\mathcal{S}^2$ norm

$$\|U\|_{\mathcal{S}_\pi^2} := \left(\max_{0 \leq i \leq N-1} \mathbb{E}|U_{t_i}|^2 \right)^{1/2}.$$

Then

$$\|Y - \widehat{\mathcal{U}}\|_{\mathcal{S}_\pi^2} \leq \|Y - Y^R\|_{\mathcal{S}_\pi^2} + \|Y^R - \bar{Y}^\pi\|_{\mathcal{S}_\pi^2} + \|\bar{Y}^\pi - \widehat{\mathcal{U}}\|_{\mathcal{S}_\pi^2}.$$

The three terms are respectively the passage from continuous to discrete reflection, the time discretisation, and the replacement of the conditional expectations by networks. For the first, [27, Lem. 4.2] gives

$$\max_j \left\| \sup_{t \in [r_j, r_{j+1}]} |Y_t - Y_t^R| \right\|_{L^2} \leq C|\pi|^{1/4},$$

improving to $C|\pi|^{1/2}$ under (R1). For the second, the system being Lipschitz, [27, Lem. 4.1] gives

$$\|Y^R - \bar{Y}^\pi\|_{L^2} \leq C(\alpha_1(N)|\pi|^{1/2} + \epsilon_1(\pi)), \quad (\alpha_1, \epsilon_1) = \begin{cases} (N^{1/4}, |\pi|^{1/4}) & \text{under (R1),} \\ (1, |\pi|^{1/2}) & \text{under (R2).} \end{cases}$$

Since $N|\pi| \leq L$,

$$\alpha_1(N)|\pi|^{1/2} = \begin{cases} N^{1/4}|\pi|^{1/2} \leq L^{1/4}|\pi|^{1/4} & \text{under (R1),} \\ |\pi|^{1/2} & \text{under (R2),} \end{cases}$$

and the first two terms together are $O(|\pi|^{1/4})$ under (R1) and $O(|\pi|^{1/2})$ under (R2).

For the martingale integrand the same route is unavailable, the corresponding estimate in [27, Lem. 4.1] carrying $\alpha_2(N) = N^{1/2}$ rather than α_1 , so that $\alpha_2(N)|\pi|^{1/2} = (N|\pi|)^{1/2}$ does not vanish. One passes instead through the family driven by X^π . The stability estimate [27, Thm. 3.2] together with (28) gives

$$\|Z - Z^e\|_{\mathcal{H}^2}^2 \leq C\|X - X^\pi\| \leq C|\pi|^{1/2},$$

then $\|Z^e - Z^{R,e}\|_{\mathcal{H}^2} \leq C|\pi|^{1/4}$ by [27, Lem. 4.2], and finally the third estimate of [27, Lem. 4.1] gives

$$\|Z^{R,e} - \bar{Z}^\pi\|_{\mathcal{H}^2} \leq C(\alpha_1(N)|\pi|^{1/2} + \epsilon_2(\pi)), \quad \epsilon_2(\pi) = |\pi|^{1/4},$$

with α_1 as above. Each of these three terms is $O(|\pi|^{1/4})$, and ϵ_2 is unchanged by (R2), which is the assertion that the estimate for the integrand does not improve under the stronger regularity.

It remains to bound the network error, the third term of the decomposition above. By (32) the system driven by g^{M_z} satisfies hypothesis (H1) of [19], and Algorithm 1 is their RDBDP scheme with min in place of max in the projection step (35). The argument of [19, proof of Thm. 4.4, from (4.41)] therefore applies verbatim, the only change being the use of

$$|\min(a, c) - \min(b, c)| \leq |a - b|, \quad a, b, c \in \mathbb{R},$$

in place of the corresponding inequality for max. That argument uses only the Lipschitz property of the driver and the identity $\arg \min_\theta \hat{L}_i(\theta) = \arg \min_\theta \tilde{L}_i(\theta)$, and is insensitive to the law of the forward process, so X may be replaced by X^π throughout. It yields

$$\max_{i=0, \dots, N-1} \mathbb{E} |\bar{Y}_{t_i}^\pi - \hat{U}_i(X_{t_i}^\pi)|^2 \leq C \sum_{i=0}^{N-1} (N\epsilon_i^{\mathcal{N}, \tilde{u}} + \epsilon_i^{\mathcal{N}, \tilde{z}}),$$

and the corresponding bound for the integrand by the argument of [19, proof of Thm. 4.1, Step 5]. Combining the displays gives (37). $\square$

Remark 5.5. The constant C in (37) depends on the Lipschitz constant of g^{M_z} , hence by (32) on M_z , which involves exponential terms in the data. The estimate is therefore qualitatively sharp but numerically weak. An alternative, developed for unreflected quadratic BSDEs in [6], is to let the truncation radius grow with the grid rather than fixing it, balancing a nonzero truncation error against a slowly growing Lipschitz constant and recovering a rate $|\pi|^{1-\eta}$ for every $\eta > 0$ with a controlled constant. Extending that construction to the reflected deep backward scheme is left for future work, as is the case of state-dependent σ (Remark 5.3).

Remark 5.6. The discretisation estimates are obtained in [27] for a scheme in which the conditional expectations are computed exactly, and the network error estimate in [19] for a Lipschitz driver. The contribution of this section is the combination of the two, the treatment of the upper obstacle, and the application to the drivers of Section 6.

6 Examples

The purpose of this section is to illustrate how different specifications of the reflected BSDE driver correspond to different financial interpretations of ambiguity and risk evaluation. In the

reflected setting, the driver determines the nonlinear risk assessment during the continuation region, while the obstacle represents the immediate liquidation or exercise value. Thus each example below should be read as a particular financial mechanism generating a dynamic stopping risk measure.

From a risk measure perspective, the variable Y represents the current risk adjusted value of the future position, whereas Z captures the exposure of this value to Brownian market uncertainty. A linear term in Y naturally corresponds to discounting, or equivalently to the time value of future cash flows. A quadratic term in Z corresponds to entropic model ambiguity, exponential utility valuation, or robustness with respect to changes of probability measure.

The first example isolates ambiguity in the discount factor and is related to cash-subadditive risk measures. The second example recalls the classical discounted entropic driver. The third example combines both mechanisms by allowing uncertainty in the discount rate and in the entropic ambiguity parameter. Finally, the American put example shows how these drivers enter a reflected BSDE with an economically meaningful stopping boundary.

6.1 Bounded discount rate ambiguity

Let the discount rate belong to the bounded interval

$$\beta_t \in [\underline{\beta}, \bar{\beta}], \quad 0 \leq \underline{\beta} \leq \bar{\beta}.$$

For $t \leq \tau$, define

$$\rho_{t,\tau}(\xi_\tau) := \sup_{\beta \in [\underline{\beta}, \bar{\beta}]} \mathbb{E} \left[e^{-\int_t^\tau \beta_s ds} (-\xi_\tau) \mid \mathcal{F}_t \right].$$

This is a pure discount-ambiguity risk measure. The probability measure is fixed, and the adversary only chooses the discount rate.

The associated reflected BSDE driver is

$$g(t, y, z) = \sup_{\beta \in [\underline{\beta}, \bar{\beta}]} \{-\beta y\}.$$

For simplicity and to obtain an explicit benchmark, we take the ambiguity bounds $\underline{\beta}$ and $\bar{\beta}$ to be constant. More generally, one may consider bounded progressively measurable processes $\underline{\beta}_t$ and $\bar{\beta}_t$, with $0 \leq \underline{\beta}_t \leq \bar{\beta}_t$, without changing the basic supremum structure of the driver. Equivalently,

$$g(t, y, z) = -\bar{\beta} y \mathbf{1}_{\{y < 0\}} - \underline{\beta} y \mathbf{1}_{\{y \geq 0\}}.$$

The stopping value

$$Y_t = \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \rho_{t,\tau}(\xi_\tau)$$

is represented by the upper reflected BSDE

$$Y_t = -\xi_T + \int_t^T \sup_{\beta \in [\underline{\beta}, \bar{\beta}]} \{-\beta Y_s\} ds - \int_t^T Z_s dW_s - (K_T - K_t),$$

with

$$Y_t \leq -\xi_t, \quad \int_0^T (Y_t + \xi_t) dK_t = 0.$$

The optimal stopping time is

$$\tau_t^* = \inf\{s \geq t : Y_s = -\xi_s\}.$$

In the liability region, where $Y_s \leq 0$, the worst case discount rate is $\bar{\beta}$. Hence the equation reduces locally to

$$Y_t = -\xi_T + \int_t^T (-\bar{\beta} Y_s) ds - \int_t^T Z_s dW_s - (K_T - K_t).$$

If the payoff process is deterministic, continuous, and nonnegative, then

$$\rho_{t,s}(\xi_s) = -e^{-\bar{\beta}(s-t)}\xi_s, \quad s \in [t, T],$$

and therefore

$$Y_t = - \sup_{s \in [t, T]} \left\{ e^{-\bar{\beta}(s-t)}\xi_s \right\}.$$

An optimal stopping time is any first maximizer of the discounted payoff:

$$\tau_t^* = \inf \left\{ s \in [t, T] : e^{-\bar{\beta}(s-t)}\xi_s = \sup_{u \in [t, T]} e^{-\bar{\beta}(u-t)}\xi_u \right\}.$$

6.2 Linear discounting and change of measure

We next consider an affine driver combining deterministic discounting with a linear change-of-measure component. Let

$$g(t, y, z) = -\beta y + \gamma \cdot z, \quad \beta \geq 0, \quad \gamma \in \mathbb{R}^d.$$

For simplicity, β and γ are taken to be constant in this benchmark. More generally, the same representation extends to suitable bounded progressively measurable processes β_t and γ_t , provided the corresponding stochastic exponential is a true martingale.

Since the driver is affine in (y, z) , Theorem 3.2-(e) implies that the associated stopping operator φ_t is concave.

To obtain an explicit representation of the corresponding dynamic risk evaluation, define the probability measure $\mathbb{Q}^\gamma$ by

$$\left. \frac{d\mathbb{Q}^\gamma}{d\mathbb{P}} \right|_{\mathcal{F}_s} = \mathcal{E} \left(\int_0^s \gamma \cdot dW_u \right)_s = \exp \left\{ \int_0^s \gamma \cdot dW_u - \frac{1}{2} \int_0^s |\gamma|^2 du \right\}.$$

Since γ is constant, Novikov's condition is automatically satisfied, and hence the stochastic exponential is a true martingale. By Girsanov's theorem, the process

$$W_s^\gamma := W_s - \int_0^s \gamma du = W_s - \gamma s$$

is a d -dimensional Brownian motion under $\mathbb{Q}^\gamma$.

Indeed, for a fixed horizon $s \in [t, T]$, the BSDE associated with the dynamic risk evaluation is

$$Y_r = -\xi_s + \int_r^s (-\beta Y_u + \gamma \cdot Z_u) du - \int_r^s Z_u dW_u, \quad r \in [t, s].$$

In differential form,

$$dY_r = \beta Y_r dr - \gamma \cdot Z_r dr + Z_r dW_r.$$

Since

$$dW_r = dW_r^\gamma + \gamma dr,$$

the terms involving $\gamma \cdot Z_r$ cancel, and therefore

$$dY_r = \beta Y_r dr + Z_r dW_r^\gamma.$$

Multiplying by the discount factor $e^{-\beta(r-t)}$ gives

$$d \left(e^{-\beta(r-t)} Y_r \right) = e^{-\beta(r-t)} Z_r dW_r^\gamma.$$

Hence the discounted process is a $\mathbb{Q}^\gamma$ -martingale. Using the terminal condition $Y_s = -\xi_s$, we obtain

$$\rho_{t,s}(\xi_s) = Y_t = \mathbb{E}^{\mathbb{Q}^\gamma} \left[e^{-\beta(s-t)}(-\xi_s) \mid \mathcal{F}_t \right], \quad \xi_s \in L^\infty(\mathcal{F}_s).$$

Consequently, the corresponding optimal stopping risk functional admits the representation

$$\varphi_t(\xi) = \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \rho_{t,\tau}(\xi_\tau) = \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \mathbb{E}^{\mathbb{Q}^\gamma} \left[e^{-\beta(\tau-t)}(-\xi_\tau) \mid \mathcal{F}_t \right].$$

Thus, in this affine case, the nonlinear stopping problem reduces to a classical optimal stopping problem under the equivalent probability measure $\mathbb{Q}^\gamma$, with discount rate β .

6.3 Linear discounting with entropic model ambiguity

A classical financially meaningful driver is

$$g(t, y, z) = -\delta y + \frac{\gamma}{2}|z|^2, \quad \delta \geq 0, \quad \gamma > 0.$$

The term $-\delta y$ represents linear discounting, while $(\gamma/2)|z|^2$ is the standard entropic or model-ambiguity term.

The convex conjugate is computed as

$$g^*(t, \beta, \alpha) = \sup_{(y,z) \in \mathbb{R} \times \mathbb{R}^d} \{-\beta y - \alpha \cdot z - g(t, y, z)\}.$$

Hence

$$g^*(t, \beta, \alpha) = \sup_{y \in \mathbb{R}} \{(-\beta + \delta)y\} + \sup_{z \in \mathbb{R}^d} \left\{ -\alpha \cdot z - \frac{\gamma}{2}|z|^2 \right\}.$$

The supremum in z is attained at

$$z^* = -\frac{\alpha}{\gamma},$$

and gives

$$\sup_{z \in \mathbb{R}^d} \left\{ -\alpha \cdot z - \frac{\gamma}{2}|z|^2 \right\} = \frac{1}{2\gamma}|\alpha|^2.$$

Therefore

$$g^*(t, \beta, \alpha) = \begin{cases} \frac{1}{2\gamma}|\alpha|^2, & \beta = \delta, \\ +\infty, & \beta \neq \delta. \end{cases}$$

Thus this model has entropic ambiguity in the probability measure, while the discount rate is fixed at δ . It serves as the benchmark case for the paired ambiguity model below.

6.4 Entropic model ambiguity with uncertain discounting

We now allow simultaneous ambiguity in the discount rate and in the entropic ambiguity parameter. Let

$$g(t, y, z) = \sup_{(\delta, \gamma) \in C} \left\{ -\delta y + \frac{\gamma}{2}|z|^2 \right\},$$

where δ represents the discount rate and γ represents the strength of entropic model ambiguity.

A separable ambiguity set is

$$C = [\underline{\delta}, \bar{\delta}] \times [\underline{\gamma}, \bar{\gamma}],$$

with

$$0 \leq \underline{\delta} \leq \bar{\delta}, \quad 0 < \underline{\gamma} \leq \bar{\gamma}.$$

In this case,

$$g(t, y, z) = \sup_{\delta \in [\underline{\delta}, \bar{\delta}]} (-\delta y) + \sup_{\gamma \in [\underline{\gamma}, \bar{\gamma}]} \frac{\gamma}{2} |z|^2.$$

Therefore

$$g(t, y, z) = -\bar{\delta} y \mathbf{1}_{\{y < 0\}} - \underline{\delta} y \mathbf{1}_{\{y \geq 0\}} + \frac{\bar{\gamma}}{2} |z|^2.$$

In the liability region $y \leq 0$, this reduces to

$$g(t, y, z) = -\bar{\delta} y + \frac{\bar{\gamma}}{2} |z|^2.$$

For simplicity, we take the ambiguity bounds $\underline{\delta}, \bar{\delta}, \underline{\gamma}, \bar{\gamma}$ to be constant in this benchmark. More generally, one may consider bounded progressively measurable bounds

$$\underline{\delta}_t \leq \bar{\delta}_t, \quad \underline{\gamma}_t \leq \bar{\gamma}_t,$$

leading to the time-dependent ambiguity set

$$\mathcal{C}_t = [\underline{\delta}_t, \bar{\delta}_t] \times [\underline{\gamma}_t, \bar{\gamma}_t].$$

The constant rectangular specification considered here is sufficient to illustrate explicitly the interaction between discount rate ambiguity and entropic model ambiguity.

We deliberately use a rectangular ambiguity set, so that the two sources of ambiguity can vary independently. Indeed, the discount parameter δ and the entropic parameter γ play different roles in the driver: δ acts on the value variable y , whereas γ controls the quadratic exposure term $|z|^2$. For this reason, no direct ordering between δ and γ is imposed.

For fixed (y, z) , the worst-case parameters can be identified explicitly. Since

$$\frac{\gamma}{2} |z|^2$$

is nondecreasing in γ , the supremum over the entropic ambiguity parameter is attained at

$$\gamma^* = \bar{\gamma}.$$

Similarly,

$$\delta^*(y) = \begin{cases} \bar{\delta}, & y < 0, \\ \underline{\delta}, & y \geq 0. \end{cases}$$

Hence, in the liability region $y \leq 0$, the worst-case specification is

$$(\delta^*, \gamma^*) = (\bar{\delta}, \bar{\gamma}),$$

and the driver becomes

$$g(t, y, z) = -\bar{\delta} y + \frac{\bar{\gamma}}{2} |z|^2.$$

The associated stopping value

$$Y_t = \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \rho_{t, \tau}(\xi_\tau)$$

is represented by the upper reflected BSDE

$$Y_t = -\xi_T + \int_t^T \sup_{(\delta, \gamma) \in \mathcal{C}} \left\{ -\delta Y_s + \frac{\gamma}{2} |Z_s|^2 \right\} ds - \int_t^T Z_s dW_s - (K_T - K_t),$$

with

$$Y_t \leq -\xi_t, \quad \int_0^T (Y_t + \xi_t) dK_t = 0.$$

The optimal stopping time is

$$\tau_t^* = \inf\{s \geq t : Y_s = -\xi_s\}.$$

If the payoff is deterministic and nonnegative, then $Z \equiv 0$. In the liability region, the driver becomes

$$g(t, y, 0) = -\bar{\delta}y.$$

Consequently,

$$\rho_{t,s}(\xi_s) = -e^{-\bar{\delta}(s-t)}\xi_s,$$

and the stopping value is explicitly

$$Y_t = - \sup_{s \in [t, T]} \left\{ e^{-\bar{\delta}(s-t)} \xi_s \right\}.$$

Thus, in the deterministic case, the optimal stopping rule is to stop at the first time at which the worst case discounted payoff is maximal.

6.5 Geometric American put under entropic-discount ambiguity

We consider an American put written on a single asset whose risk neutral price S follows a geometric Brownian motion. In log-price coordinates $X_t = \log S_t$, the dynamics are

$$dX_t = \left(r - \frac{1}{2}\sigma^2 \right) dt + \sigma dW_t, \quad X_0 = \log S_0.$$

The exercise payoff is

$$\xi_t = (K - S_t)^+ = (K - e^{X_t})^+.$$

It is bounded by K and has continuous paths, hence it satisfies the standing obstacle assumptions.

Using the entropic discount ambiguity driver

$$g(t, y, z) = \sup_{(\delta, \gamma) \in C} \left\{ -\delta y + \frac{\gamma}{2} |z|^2 \right\},$$

the stopping value

$$Y_t = \operatorname{ess\,inf}_{\tau \in \mathcal{T}_t} \rho_{t,\tau}(\xi_\tau)$$

solves the upper reflected BSDE

$$Y_t = -\xi_T + \int_t^T g(s, Y_s, Z_s) ds - \int_t^T Z_s dW_s - (K_T - K_t), \quad Y_t \leq -\xi_t,$$

with

$$\int_0^T (Y_t + \xi_t) dK_t = 0.$$

For the separable ambiguity set

$$C = [\underline{\delta}, \bar{\delta}] \times [\underline{\gamma}, \bar{\gamma}],$$

and in the liability region $Y_t \leq 0$, the driver reduces to

$$g(t, Y_t, Z_t) = -\bar{\delta}Y_t + \frac{\bar{\gamma}}{2} |Z_t|^2.$$

Thus the reflected BSDE used for the American put becomes

$$Y_t = -\xi_T + \int_t^T \left(-\bar{\delta}Y_s + \frac{\bar{\gamma}}{2} |Z_s|^2 \right) ds - \int_t^T Z_s dW_s - (K_T - K_t), \quad Y_t \leq -\xi_t.$$

The initial asset level and strike may be chosen as $S_0 = 1$ and $K = 1.1$. Then the option starts slightly in the money and the initial upper obstacle is

$$-\xi_0 = -(K - S_0)^+ = -0.1.$$

This makes the stopping constraint active and visible in the numerical diagnostics.

7 Numerical Diagnostics and Results

We use the reflected deep backward dynamic programming scheme of Section 5, Algorithm 1, to solve the upper reflected BSDE (4)–(6) on three test cases:

- (1) American put under Black–Scholes ($g(y) = -ry$, no ambiguity), the standard optimal stopping benchmark for which a binomial reference price is available.
- (2) Bounded discount rate ambiguity of Section 6.1, a standard linear example under ambiguity that still falls within the Lipschitz framework.
- (3) Geometric American put under entropic discount ambiguity of Section 6.5, which introduces the superlinear (quadratic in z) driver and is the non-Lipschitz target of the convergence analysis in Section 5.

Remark 7.1. The drivers of Section 6 satisfy (H3)–(H4). They are of the form

$$g(t, x, y, z) = \sup_{(\delta, \gamma) \in \mathcal{C}} \psi(\delta, \gamma, y, z),$$

with $\mathcal{C}$ bounded, $\delta \in [\underline{\delta}, \bar{\delta}]$ and $\gamma \in [\underline{\gamma}, \bar{\gamma}]$, and are independent of (t, x) ; hence (24) holds with $L = 0$ and (H4) is vacuous for g . For $\psi(\delta, \gamma, y, z) = -\delta y + \frac{\gamma}{2}|z|^2$,

$$-\bar{\delta}|y| \leq g(y, z) \leq \bar{\delta}|y| + \frac{\bar{\gamma}}{2}|z|^2,$$

giving (23) with $M_g = \bar{\delta}$ and $\gamma = \bar{\gamma}$, while $|\sup_a F(a, \cdot) - \sup_a F(a, \cdot)| \leq \sup_a |F(a, \cdot) - F(a, \cdot)|$ gives

$$|g(y, z) - g(y', z)| \leq \sup_{\delta} \delta|y - y'| \leq \bar{\delta}|y - y'|, |g(y, z) - g(y, z')| \leq \sup_{\gamma} \frac{\gamma}{2}||z|^2 - |z'|^2| \leq \frac{\bar{\gamma}}{2}(|z| + |z'|)|z - z'|,$$

which are (25) and (26). The remaining drivers of Section 6 are special cases. Boundedness of $\mathcal{C}$ is what these estimates require.

Cases (1) and (3) use the payoff $h(x) = (K - e^x)^+$, which is bounded and Lipschitz but not differentiable at the strike. They therefore satisfy the standing assumptions (H1)–(H4) of Section 5, but not the additional regularity (R1)–(R2) under which the discretisation error in Proposition 5.4 is quantified. Case (2) satisfies all of them, and is the example for which the rate of Proposition 5.4 is available. For each case we report (i) sample paths of the value process Y_t against the obstacle $-\xi_t$ and of the control process Z_t , (ii) the empirical distribution of optimal stopping times, and (iii) the training loss at the terminal time step. Numerical verification of the relevant properties of Theorem 3.2 is collected in Section 7.4.

Common solver configuration. All experiments use the same network and training hyperparameters (Table 1). Outputs go through the upper-reflection projection (35). Implementation is in Python with PyTorch on CPU.

Architecture	3 hidden layers, 50 neurons per layer
Activation	tanh on hidden layers, identity on output
Optimizer	Adam, learning rate 10^{-3}
Batch size	$M = 2^{10} = 1024$
Time discretization	$N = 50$ steps, $\Delta t = T/N$
Base iterations per step	itr = 300
Iteration multiplier on last 3 steps	$\times 10$

Table 1: Network and training hyperparameters used in all three cases.

7.1 American put under Black–Scholes

This case is the canonical optimal stopping benchmark: the reflected BSDE of Theorem 3.1 reduces, under the affine driver $g(y) = -ry$ and the put payoff $\xi_t = (K - S_t)^+$, to the classical American put under Black–Scholes. The Cox–Ross–Rubinstein binomial price is available as a reference, which fixes the baseline accuracy of the scheme before any ambiguity is introduced.

The forward process is the GBM in log coordinates of Section 6.5; the upper obstacle is $-\xi_t = -(K - e^{X_t})^+$. The driver is recovered from Section 6 as the discount-only special case of Section 6.1 with $\underline{\beta} = \bar{\beta} = r$. Market parameters are listed in Table 2.

Initial price	S_0	1.0
Strike	K	1.1
Volatility	σ	0.2
Risk-free rate	r	0.05
Time horizon	T	1.0

Table 2: Market parameters, American put under Black–Scholes.

The Cox–Ross–Rubinstein binomial price with $N = 2000$ time steps yields $P_{\text{ref}} = 0.11973$ (the intrinsic value at $t = 0$ is $K - S_0 = 0.1$). The solver returns $-Y_0 = 0.11638$, an absolute gap of -0.0034 and a relative deviation of -2.8% .

Figure 1 collects the three diagnostics. Panel 1a displays, on the left, three realised value paths Y_t with the corresponding moving obstacle $-\xi_t$; the value process stays below the obstacle pathwise and makes contact during sufficiently in-the-money excursions. On the right, the same panel shows the realised control process Z_t , which fluctuates around $-\sigma \cdot \partial_x u$ and is mostly negative (the put has nonpositive delta in the state $X_t = \log S_t$). Panel 1b is the histogram of the realised first-contact times τ^* of Theorem 3.1 across 1024 simulated paths. Roughly 41% of paths exit strictly before T , with mean conditional stopping time $\bar{\tau}^* \approx 0.47$. The cluster of values just below T visible in the histogram is largely a consequence of the imposed terminal contact $Y_T = -\xi_T$, since the contact time detector therefore identifies paths near the terminal step. Panel 1c reports the training loss for the terminal step $n = N - 2$ on a log scale; the loss drops by more than two orders of magnitude and plateaus near 10^{-4} . The losses at the other time steps are of the same shape and reach smaller magnitudes (closer to 10^{-6}), since the conditional-expectation regression smooths the obstacle non-smoothness as it propagates backward.

7.2 Bounded discount rate ambiguity

This case applies the scheme to the driver of Section 6.1, $g(y) = \sup_{\beta \in [\underline{\beta}, \bar{\beta}]} \{-\beta y\}$, which is convex in y but no longer affine. Both branches of the sup become active only when Y visits both signs, so a nonnegative put payoff would pin $Y \leq 0$ and collapse the band to $\bar{\beta}$. We therefore take the collared obstacle

$$\xi_t = c \tanh(X_t/c), \quad dX_t = dW_t,$$

which is sign changing, and bounded by c as required by (H2). It is smooth with bounded derivatives of every order, so it satisfies (R1) and (R2) as well, and this is the one test case for which the rate in Proposition 5.4 is available. Parameters are listed in Table 3.

The solver returns $Y_0 = -0.0846$. Across interior time steps, 44.7% of (t, ω) samples have $Y_t \geq 0$, confirming that both branches of the sup are exercised. Evaluated on 8192 simulated paths, a fraction 0.54 of paths make first contact strictly before T , with mean $\bar{\tau}^* \approx 0.43$.

Figure 2 reports the diagnostics. Panel 2a shows, on the left, the value process Y_t together with the moving obstacle $-\xi_t$ for three sample paths; the value stays below the obstacle and

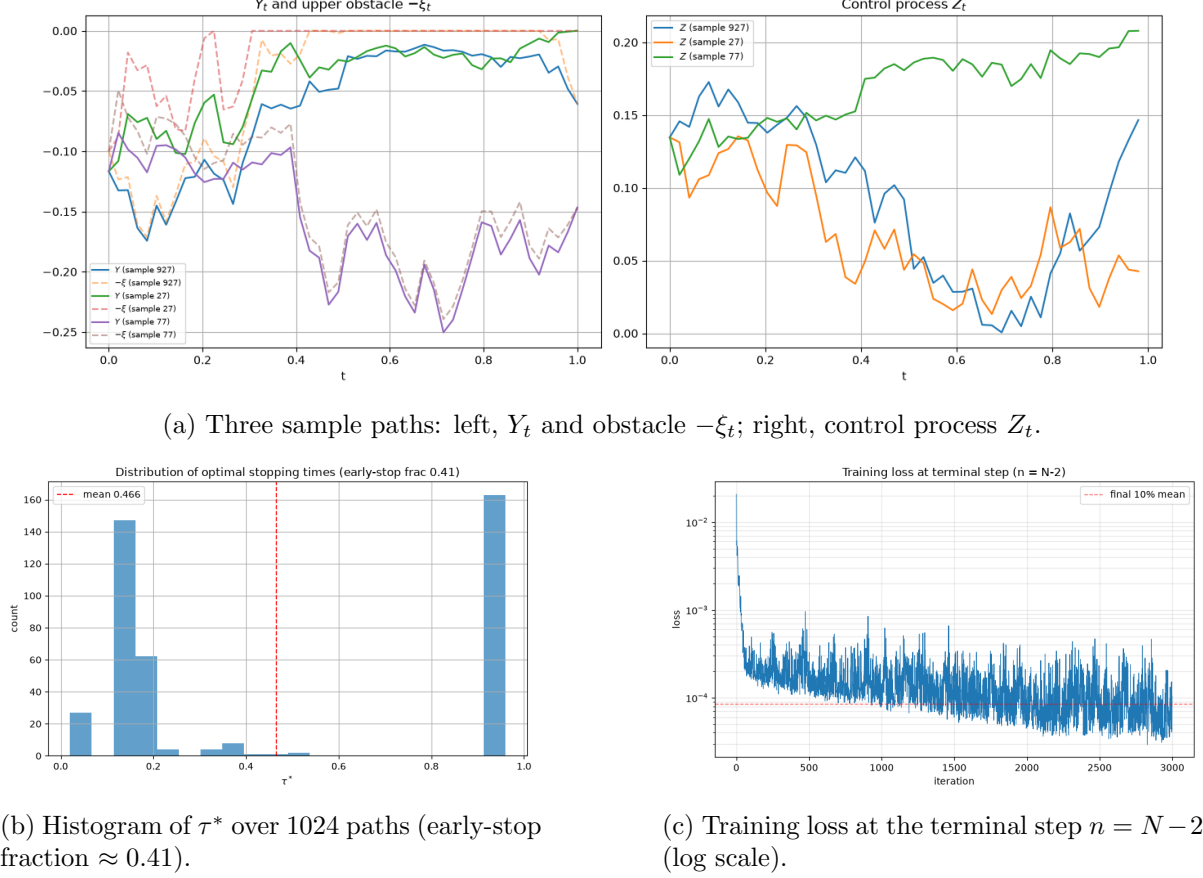


Figure 1: Black-Scholes American put: diagnostics from a single trained run.

Forward	$dX_t = dW_t$	$X_0 = 0$
Obstacle process	$\xi_t = c \tanh(X_t/c)$	sign-changing, $ \xi \leq c$
Collar level	c	1.0
Lower discount bound	$\underline{\beta}$	0.0
Upper discount bound	$\overline{\beta}$	0.1
Time horizon	T	1.0

Table 3: Forward, obstacle and discount band, Section 6.1 case.

makes contact on excursions where X_t becomes sufficiently negative. The collar flattens the obstacle for large $|X_t|$, so contact occurs earlier than it would for an unbounded payoff. On the right, the same panel shows the control process Z_t , which carries the trend of the sign-changing obstacle and is damped where the collar saturates. Panel 2b is the histogram of τ^* over the paths making contact before T , the terminal step being excluded by construction. The mass is concentrated in the interior of $[0, T]$, rising to a peak around $t = 0.5$ and decaying thereafter, with mean 0.43 and median 0.52. Panel 2c reports the terminal-step training loss; its order of magnitude and shape match Figure 1c, and the losses at earlier steps are smaller still. The band sup-structure does not destabilise the recursion.

7.3 Geometric American put under entropic-discount ambiguity

This case is the non-Lipschitz target of the convergence analysis in Section 5: the driver of Section 6.5 carries the quadratic in z term $(\overline{\gamma}/2)|z|^2$ that lies outside the Lipschitz framework and motivates the discrete reflection and truncation analysis of Section 5. We demonstrate



Figure 2: Bounded discount rate ambiguity (Section 6.1): diagnostics from a single trained run.

that the scheme, combined with truncation and the upper reflection projection, handles the superlinear driver and that the entropic ambiguity has a measurable effect on the conservative put price relative to Section 7.1.

In the liability region $Y \leq 0$ the driver of Section 6.5 reduces to $g(y, z) = -\bar{\delta}y + (\bar{\gamma}/2)|z|^2$. We take $\bar{\delta} = r$, so the worst case discount equals the risk-free rate, and $\bar{\gamma} = 5$ (motivated below). The forward dynamics, payoff and market parameters are inherited from Table 2; the additional ambiguity parameters are in Table 4.

Worst-case discount	$\bar{\delta}$	0.05
Entropic radius	$\bar{\gamma}$	5.0

Table 4: Ambiguity parameters, Section 6.5 case.

A sweep over $\bar{\gamma} \in \{0.5, 1, 2, 5, 10\}$ at fixed market parameters yields $-Y_0$ equal to 0.115, 0.115, 0.112, 0.109 and 0.107 respectively. Below $\bar{\gamma} \approx 2$ the entropic effect is buried in the optimal stopping bias; above $\bar{\gamma} \approx 10$ the conservative price approaches the intrinsic value $K - S_0 = 0.1$ and the continuation region collapses. The value $\bar{\gamma} = 5$ keeps the price clearly above intrinsic while the quadratic term contributes meaningfully. At $\bar{\gamma} = 5$ the solver returns $-Y_0 = 0.10881$, a -6.5% shift from the Black-Scholes baseline of Section 7.1.

Figure 3 reports the diagnostics. Panel 3a shows, on the left, the value process Y_t with the moving obstacle $-\xi_t = -(K - S_t)^+$ for three sample paths; the obstacle constraint is respected without measurable violation. On the right, the same panel shows the realised control process Z_t , which is the quantity penalised quadratically by the entropic driver. Panel 3b is the histogram of τ^* . Comparing with Figure 1b, two changes are visible. First, the cluster just below T remains

as a consequence of the imposed terminal contact $Y_T = -\xi_T$. Second, the rest of the distribution shifts toward earlier times: the early stop fraction rises from 0.41 in Section 7.1 to 0.82 here. This empirical shift is consistent with the increased contribution of the quadratic z term in the continuation region, although the experiment does not by itself establish a comparative statics result for the optimal stopping boundary. Panel 3c reports the terminal-step training loss; it has the same shape and the same order of magnitude as in Figures 1c–2c, with losses at the earlier time steps smaller by an order of magnitude. The superlinear driver therefore does not destabilise training under the hyperparameters of Table 1.

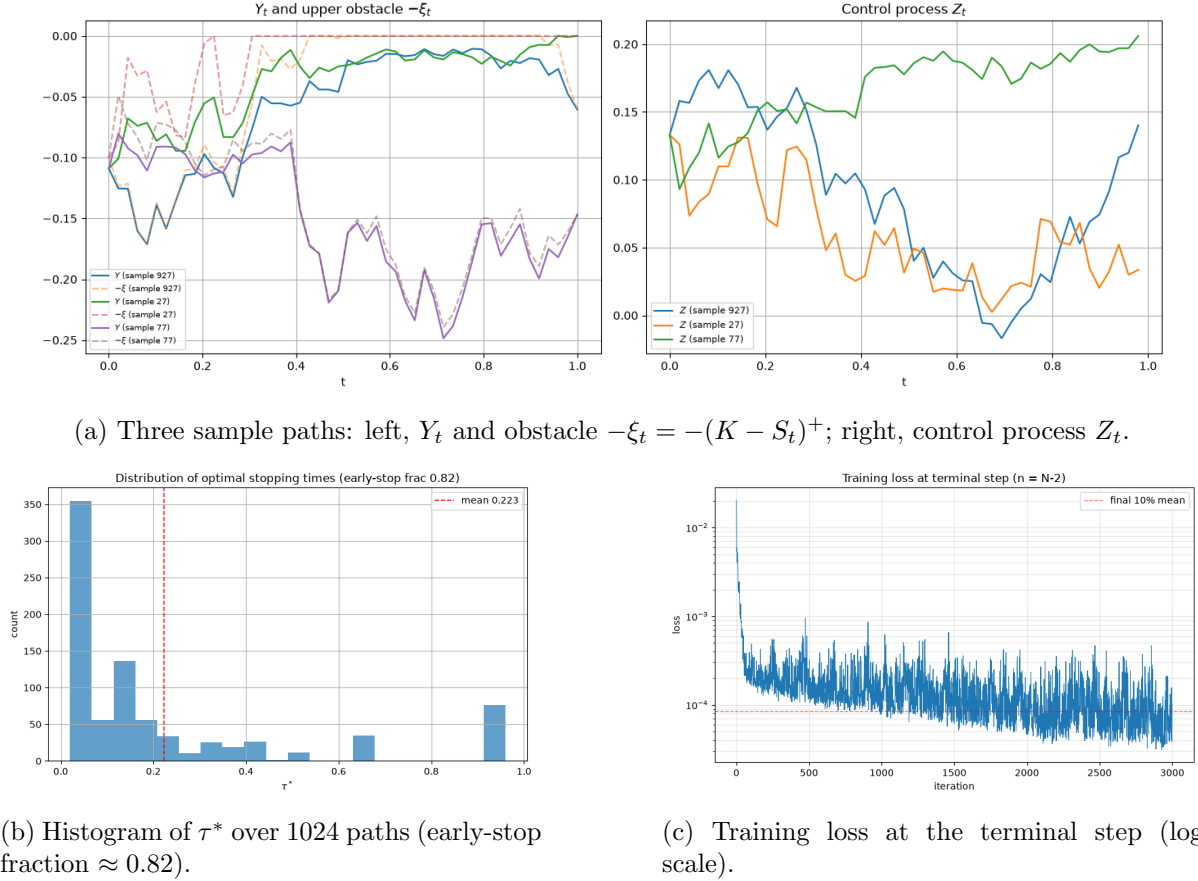


Figure 3: Entropic-discount American put (Section 6.5): diagnostics from a single trained run at $\bar{\gamma} = 5$.

7.4 Numerical verification of Theorem 3.2

For each of the cases of Section 7.1–7.3 we test on the trained solver the three properties of Theorem 3.2 that can be read off from the value $\varphi_0(\xi) = Y_0$:

- (a) $\varphi_0(\xi + m) \leq \varphi_0(\xi)$ for a constant shift $m = 0.2$;
- (b) $\varphi_0(\xi + m) \geq \varphi_0(\xi) - m$ for the same shift;
- (e) $\varphi_0(\frac{1}{2}\xi^1 + \frac{1}{2}\xi^2) \geq \frac{1}{2}\varphi_0(\xi^1) + \frac{1}{2}\varphi_0(\xi^2)$ for a basket of two obstacles.

Under the standing assumptions of Section 2.2, property (a) holds for every admissible driver, while property (b) holds for the drivers considered here because they are decreasing in y . Both properties must therefore be reproduced by the solver in all three cases. Concavity (e), on the other hand, is established only when the driver is affine in (y, z) . This is the situation of

Section 7.1, where $g(y) = -ry$; the inequality (e) is therefore expected. In Section 7.2 and Section 7.3 the driver is convex but not affine, so Theorem 3.2 makes no prediction about the sign of $\varphi_0(\frac{1}{2}\xi^1 + \frac{1}{2}\xi^2) - \frac{1}{2}[\varphi_0(\xi^1) + \varphi_0(\xi^2)]$, and the corresponding entry of the table is reported only as an empirical observation on the chosen basket.

For each case we train four networks, on the base obstacle and on its shifted, alternate, and basketed versions, at the hyperparameters of Table 1. We say that an inequality is satisfied numerically when the corresponding signed margin (defined in the caption of Table 5) is at least -0.02 ; this threshold is twice the variation observed across independent training replicates. The results are collected in Table 5.

The two unconditional inequalities (a) and (b) are satisfied in every case. Concavity (e) holds in Section 7.1, in agreement with Theorem 3.2(e). In Sections 7.2 and 7.3 the signed margin is within the numerical tolerance, so the inequality is neither clearly satisfied nor clearly violated on the chosen baskets. This is consistent with Theorem 3.2, which asserts concavity only for affine drivers and makes no prediction here; the experiments neither confirm nor refute it for the two non-affine cases.

Case	(a)	(b)	(e)
Section 7.1	+0.208	+0.012	+0.0044
Section 7.2	+0.212	+0.008	-0.009
Section 7.3	+0.214	+0.006	+0.011

Table 5: Signed margins of the three inequalities of Theorem 3.2 at $m = 0.2$ and $\lambda = \frac{1}{2}$. The quantities tabulated are $\varphi_0(\xi) - \varphi_0(\xi + m)$ for (a), $\varphi_0(\xi + m) - \varphi_0(\xi) + m$ for (b), and $\varphi_0(\frac{1}{2}\xi^1 + \frac{1}{2}\xi^2) - \frac{1}{2}[\varphi_0(\xi^1) + \varphi_0(\xi^2)]$ for (e). The numerical tolerance is 0.02. Baskets: Section 7.1 and Section 7.3 use puts with strikes $K_1 = 1.1$, $K_2 = 0.9$; Section 7.2 uses $\xi_t^1 = c \tanh(X_t/c)$ and $\xi_t^2 = c \tanh((X_t^2 - 0.5)/c)$, the collar of Section 7.2 applied to both.

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